

$$\text{STOPGATE}(D_t, G_t) = \text{Quality}_t \wedge \text{Stability}_t \wedge \text{Coverage}_t \wedge \text{Risk}_t$$

$$\text{Quality}_t = q(A_t) \geq q_{\min} \wedge \min_{c \in \{\text{H,L,N}\}} q_c(A_t) \geq q_{\min}^{\text{class}} \wedge \min_{r \in \mathcal{R}} q_r(A_t) \geq q_{\min}^{\text{region}}$$

$$\text{Stability}_t = \max_{j \in [t-K+1, t]} \Delta q_j \leq \varepsilon \wedge \text{edits editorial only} \wedge \text{no unresolved concept revision}$$

$$\text{Coverage}_t = \forall r \in \mathcal{R} : n_r(D_t) \geq n_{\min}(r) \vee \text{an accepted scarcity note}$$

$$\text{Risk}_t = \forall \rho \in \text{ledger} : \text{severity}(\rho) = \text{high} \Rightarrow \rho \text{ has an owner, a mitigation, or a recorded acceptance}$$

$\mathcal{R} = \{\text{H, L, N, HL, LN, HN, HLN}\}$ are the seven diagnostic regions. A_t is QD2's comparable audit sample, never a challenge batch, and Δq_j is the audit gain at round j . K , ε and every min threshold are project configuration fixed before the sequence is read (QD4 §2.1). A gate that fails does not lower the bar: it reopens the round.

Figure 1: The stopping rule as a conjunction. Calibration stops only when all four gates hold at once, so a strong overall score cannot buy a failed class, an unmeasured region, or an unowned high-severity risk. Every threshold is project configuration fixed before the sequence is read, and a failed gate reopens the round rather than lowering the bar.